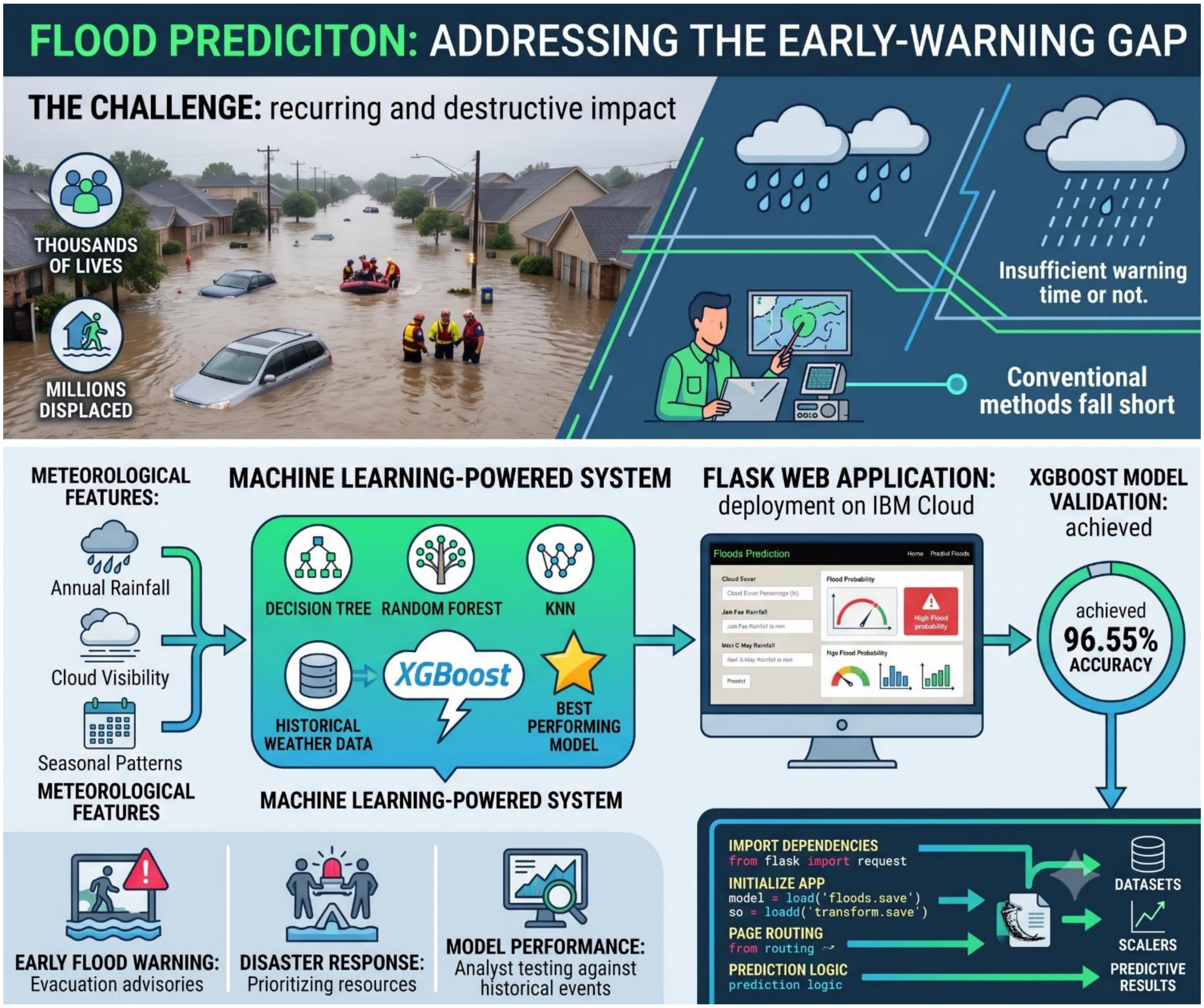

Flood disasters occur frequently due to excessive rainfall, changing climatic conditions, and environmental factors. Conventional forecasting methods often fail to provide timely and accurate flood warnings, reducing the effectiveness of evacuation and disaster response efforts.

The objective of this project is to develop a machine learning model capable of predicting flood occurrences using meteorological data, thereby improving early warning systems and minimizing the impact of floods.

Historical Weather Dataset

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▼

Data Preprocessing

│
▼

Feature Selection

│
▼

Model Training
(Decision Tree, Random Forest,
KNN, XGBoost)

│
▼

Model Evaluation
↓
Best Model (XGBoost)
↓
Model Saving (.save/.pkl)
↓
Flask Web Application
↓
User Inputs Weather Data
↓
Flood Prediction
↓
Risk Level Display

Floods are one of the most destructive natural disasters worldwide, causing significant loss of life, damage to infrastructure, and displacement of millions of people every year. Traditional flood forecasting systems often rely on meteorological observations and manual analysis, which may not provide sufficient warning time for effective disaster management. To overcome these limitations, Machine Learning (ML) techniques can be used to analyze historical weather data and predict flood occurrences with higher accuracy and speed.

This project presents a **Machine Learning-based Flood Prediction System** that utilizes historical meteorological data to predict the probability of floods. The system is deployed as a **Flask Web Application** on **IBM Cloud**, enabling users to obtain real-time flood predictions through an interactive interface.

Objectives

The major objectives of this project are:

- Develop an intelligent flood prediction system.
- Analyze historical weather data.
- Compare multiple Machine Learning algorithms.
- Select the best-performing model.
- Deploy the model as a web application.
- Provide early flood warning predictions.

XGBoost Algorithm

Extreme Gradient Boosting (XGBoost) is an advanced ensemble learning algorithm that combines multiple weak decision trees to create a strong predictive model. It uses gradient boosting techniques to minimize prediction errors while improving speed and accuracy.